

Nonlinear Model Predictive Control of Position and Attitude in a Hexacopter with Three Failed Rotors[★]

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Abstract: We applied nonlinear model predictive control (NMPC) to an unmanned aerial vehicle (a hexacopter) with three failed rotors out of the six rotors available. We considered three cases of failure: a case in which three rotors in alternate positions stop, a case in which two adjacent rotors and a separate rotor stop, and a case in which three adjacent rotors stop. Simulation results show that, by applying NMPC, the position and attitude of the hexacopter can be controlled from an initial state that significantly deviates from a state of equilibrium in consideration of constraints. Moreover, we simulated a case in which there was no static equilibrium state and succeeded in controlling the hexacopter's position.

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1. INTRODUCTION

Multirotor unmanned aerial vehicles (UAVs) have become very popular (Thompson (2015)). It is expected that they will be useful in the fields of agriculture (Rasmussen et al. (2013)), logistics (Fink et al. (2011)), disaster relief (Greer et al. (2002)), surveying (Murphy et al. (2008)), biological animal research (Selby et al. (2011)), infrastructure inspection (Nonami et al. (2016)), and so on.

However, UAVs also have a risk of accidents, such as crashes or collisions. In particular, a UAV crash can result in a serious accident when it is flying over people. Therefore, UAVs should be controlled so as not to crash even in the case of faults or failure in system components. One possible cause of UAV accidents is failure of rotors because the specified equilibrium may no longer be stabilized with the thrusts generated by the remaining rotors. To cope with such situations, it is necessary to develop a control method for a UAV on the basis of their suffering failed rotors.

Various methods of fault-tolerant control have been proposed for UAVs: sliding mode control (Sharifi et al. (2010)), gain-scheduled PID control (Milhim et al. (2010) and Sadeghzadeh et al. (2012)), PD control (Du et al. (2014)), model predictive control (Yu et al. (2013)), and LQ control (Muller et al. (2014) and Hirai (2017)). However, Sharifi et al. (2010), Milhim et al. (2010), Sadeghzadeh et al. (2012), and Yu et al. (2013) considered a reduction of thrusts in rotors rather than a complete stop, and Du et al. (2014) considered a case in which

only one rotor stopped. On the other hand, Muller et al. (2014) and Hirai (2017) considered some cases in which one, two, or three rotors stopped, but they used linearized models for control design. In some cases, it is not a trivial task to find an equilibrium, and, in any case, linearized models are invalid if the state deviates significantly from a state of equilibrium. Moreover, Hirai (2017) gave priority to altitude control and did not achieve stabilization of the horizontal position.

As preceding research related to this, stabilization control of a quadcopter or a hexacopter when a part of the rotor stops by applying LQ control has been studied (Muller et al. (2014) and Hirai (2017)). However, the UAV model of these research studies was a linear approximation. Therefore, a UAV with such a model might not be able to stabilize its attitude if it is difficult to find an equilibrium or it is significantly deviated from an equilibrium state. Consequently, errors in the model may degrade control performance. Especially, in the latter research (Hirai (2017)), position control in the horizontal direction was not achieved because attitude control was a priority. In this research, we describe a UAV with six rotors (a hexacopter) by a nonlinear model instead of the linearized models used in previous research. We considered nonlinear model predictive control (NMPC) for the hexacopter in the case of having three failed rotors out of the six available. In particular, we simultaneously controlled the position and attitude of the hexacopter in consideration of the constraints of its thrusts. We controlled five degrees of freedom except for the yaw direction with three rotors. We show that NMPC successfully controlled the hexacopter, even if it was difficult or impossible to design linear controllers.

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2. MODELING

2.1 State Equation of a Hexacopter

Our hexacopter was based on the configuration depicted in Fig. 1. To derive the equation of motion, we referred to Nonami et al. (2010). We numbered the rotors and set the

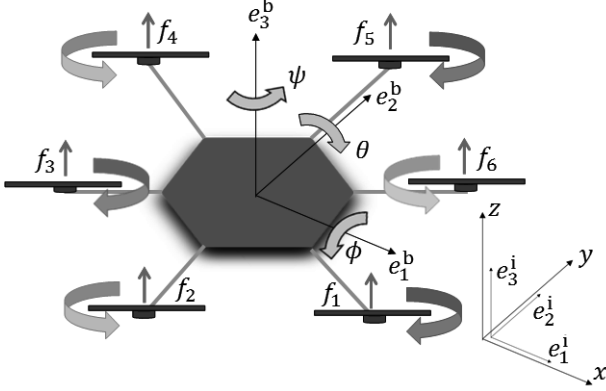


Fig. 1. Hexacopter configuration and coordinate axes

coordinate axes, as shown in Fig. 1. The coordinate axes e_j^i and e_j^b ($j = 1, 2, 3$) denote the inertial frame and body-fixed frame, respectively. The state vector and the control input vector of the hexacopter are defined as follows:

$$x_d = (\xi^T \eta^T \dot{\xi}^T \dot{\eta}^T)^T \in \mathbb{R}^{12}, \quad u_d = (f_1 \cdots f_6)^T \in \mathbb{R}^6, \quad (1)$$

where $\xi = (x \ y \ z)^T \in \mathbb{R}^3$ is the position of the hexacopter, $\eta = (\phi \ \theta \ \psi)^T \in \mathbb{R}^3$ is the vector of the ZYX Euler angles in the inertial frame, and $f_1, \dots, f_6$ are the six independent thrusts generated by the six rotors. The sum of the thrusts, f_{all} , and the torque, τ , can be written as $f_{all} = s u_d \in \mathbb{R}$ and $\tau = T u_d - D \dot{\eta} \in \mathbb{R}^3$, respectively, where $s = (1 \cdots 1) \in \mathbb{R}^{1 \times 6}$, and the matrices, $T \in \mathbb{R}^{3 \times 6}$ and $D \in \mathbb{R}^{3 \times 3}$, are defined as follows:

$$T = \begin{pmatrix} -\frac{1}{2}l & -l & -\frac{1}{2}l & \frac{1}{2}l & l & \frac{1}{2}l \\ -\frac{\sqrt{3}}{2}l & 0 & \frac{\sqrt{3}}{2}l & \frac{\sqrt{3}}{2}l & 0 & -\frac{\sqrt{3}}{2}l \\ k & -k & k & -k & k & -k \end{pmatrix}, \quad (2)$$

$$D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \gamma \end{pmatrix}, \quad (3)$$

where l is the distance from the hexacopter mass center to each rotor, k is a coefficient representing the relation between the thrust and the reaction torque, and γ is a coefficient of air resistance.

Accordingly, the state equation of the hexacopter can be described as

$$\dot{x}_d = \begin{pmatrix} \dot{\xi} \\ \dot{\eta} \\ \frac{s u_d}{m} R e_3^i - g e_3^i \\ M(\eta)^{-1} (\Psi(\eta)^T (T u_d - D \dot{\eta}) - C(\eta, \dot{\eta}) \dot{\eta}) \end{pmatrix} =: f_d(x_d, u_d), \quad (4)$$

where m is the hexacopter mass, g is gravitational acceleration, and $R \in \mathbb{R}^{3 \times 3}$ is a rotation matrix defined as:

$$R = R_\psi R_\theta R_\phi = \begin{pmatrix} c\theta c\psi & s\phi s\theta c\psi - c\phi s\psi & c\phi s\theta c\psi + s\phi s\psi \\ c\theta s\psi & s\phi s\theta s\psi + c\phi c\psi & c\phi s\theta s\psi - s\phi c\psi \\ -s\theta & s\phi c\theta & c\phi c\theta \end{pmatrix}. \quad (5)$$

Note that $s(\cdot)$ and $c(\cdot)$ are abbreviations for $\sin(\cdot)$ and $\cos(\cdot)$, respectively. $M(\eta) = \Psi^T(\eta) J \Psi(\eta)$ ($\theta \neq n\pi/2; n = \pm 1, \pm 2, \dots$) is a positive-definite inertia matrix, where $J \in \mathbb{R}^{3 \times 3}$ is the inertia matrix of the hexacopter. The matrix $\Psi(\eta) \in \mathbb{R}^{3 \times 3}$ is given by:

$$\Psi(\eta) = \begin{pmatrix} 1 & 0 & -\sin \theta \\ 0 & \cos \phi & \cos \theta \sin \phi \\ 0 & -\sin \phi & \cos \theta \cos \phi \end{pmatrix}. \quad (6)$$

Finally, $C(\eta, \dot{\eta})$ is a Coriolis term

$$C(\eta, \dot{\eta}) = \Psi(\eta)^T J \dot{\Psi}(\eta) + \Psi(\eta)^T \nu(\Psi(\eta) \dot{\eta}) J \Psi(\eta). \quad (7)$$

In this equation, $\nu(x)$ is a skew-symmetric matrix defined by $\nu(x)y := x \times y$ for any vector, $y \in \mathbb{R}^3$.

2.2 Cases of Three Failed Rotors

The objective of this research was to control the position and attitude of a hexacopter with three failed rotors, and it was necessary to consider all possible cases of failure. It was assumed that the loss of thrusts in failed rotors could be detected by some means.

There are essentially three cases of failure of three rotors due to symmetry, as shown in Figs. 2 to 4. In the first case, three rotors in alternate positions stop (Case 1), in the second case, two adjacent rotors and a separate rotor stop (Case 2), and in the third case, three adjacent rotors stop (Case 3). First, we approached Case 1, in which three rotors in alternate positions stop (Fig. 2). For modeling, it was assumed that all rotors were on the vertexes of a regular hexagon, as shown in Fig. 1. Therefore, it sufficed to consider a case in which rotors No. 2, No. 4, and No. 6 stopped. The control input u_d in equation (4) is expressed as

$$u_d = (f_1 \ 0 \ f_3 \ 0 \ f_5 \ 0)^T \in \mathbb{R}^6. \quad (8)$$

Since all the remaining rotors rotated in the same direction, it was inevitable that the rotation of yaw direction occurred.

Next, we approached Case 2, in which two adjacent rotors and a separate rotor stop (Fig. 3). In this case, it sufficed to consider a case where rotors No. 3, No. 5, and No. 6 stopped. The control input u_d in equation (4) is expressed as

$$u_d = (f_1 \ f_2 \ 0 \ f_4 \ 0 \ 0)^T \in \mathbb{R}^6. \quad (9)$$

Note that, since rotor No. 2 could not generate a negative thrust, the rotational motion around the axis connecting

rotors No. 1 and No. 4 could not be stabilized by any linear feedback controller.

Finally, we approached Case 3, in which three adjacent rotors stop (Fig. 4). In this case, it sufficed to consider a case in which rotors No. 4, No. 5, and No. 6 stopped. The control input u_d in equation (4) is expressed as

$$u_d = (f_1 \ f_2 \ f_3 \ 0 \ 0 \ 0)^T \in \mathbb{R}^6. \quad (10)$$

As can be seen from the configuration of the rotors, there was no static equilibrium, and it was not trivial task to find an equilibrium with nonzero angular velocity.

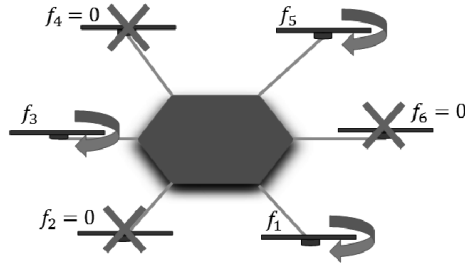


Fig. 2. Case 1: three rotors in alternate positions stop

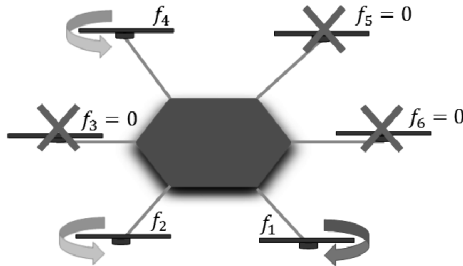


Fig. 3. Case 2: two adjacent rotors and a separated rotor stop

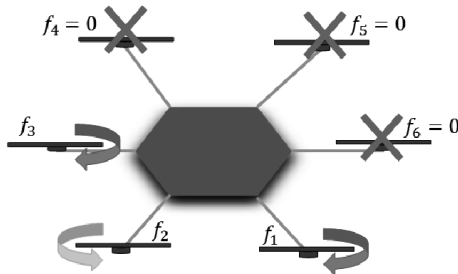


Fig. 4. Case 3: three adjacent rotors stop

3. CONTROL METHODS

The control problem of a hexacopter with three failed rotors can be considered as an optimal control problem.

NMPC is used to perform feedback control by numerically solving an optimal control problem. We used a real-time numerical algorithm called C/GMRES (Ohtsuka (2004)) to solve the NMPC problem. There are a lot of numerical algorithms for NMPC (Diehl et al. (2009), Wolf et al. (2016), and Ohtsuka (2017)). Any of those algorithms could be applied to our NMPC problem as long as real-time optimization is achieved. In this section, we summarize the NMPC and formulate the control problem of the hexacopter.

3.1 Nonlinear Model Predictive Control

We consider a general nonlinear system governed by the state equation,

$$\dot{x}(t) = f(x(t), u(t), p(t)), \quad (11)$$

where $x(t)$ is the state vector, $u(t)$ is the control input vector, and $p(t)$ is the vector of given time-varying parameters. Moreover, an equality condition is imposed as $C(x(t), u(t), p(t)) = 0$. An inequality constraint can be transformed into an equality constraint by introducing a dummy input. We consider a performance index, J , with a receding finite horizon, represented as follows:

$$J = \varphi(x(t+T), p(t+T)) + \int_t^{t+T} L(x(\tau), u(\tau), p(\tau)) d\tau. \quad (12)$$

For each time t , $x(\tau)$, $u(\tau)$ and $p(\tau)$ ($t \leq \tau \leq t+T$) respectively denote the state vector, control input vector, and vector of the given time-varying parameters on the finite horizon. Note that these vectors predicted on the horizon do not necessarily coincide with the actual state, control input, or time-varying parameter.

By choosing the appropriate functions, φ and L , for the performance index, we can formulate various control problems. The optimal control $u_{opt}(\tau)$ ($t \leq \tau \leq t+T$) minimizing J is determined as a time function within the horizon. Accordingly, the actual control input $u(t)$ at time t is given by the initial value of $u_{opt}(\tau)$, which acts as a state feedback control. This feedback control method is called NMPC or nonlinear receding-horizon control.

3.2 Control Methods of a Hexacopter with Failed Rotors

Here, we formulated the control problem of the hexacopter. We employed the nonlinear model of the hexacopter expressed by the state equation (4). The thrust generated in each rotor was constrained as follows:

$$u_{\min} = 0.144 \leq u_i \leq 6 = u_{\max} \text{ [N]}. \quad (13)$$

Then, dummy inputs u'_i ($i = 1, \dots, 6$) were introduced to transform this inequality into an equality as follows:

$$C_i = \left(u_i - \frac{u_{\min} + u_{\max}}{2} \right)^2 + u_i'^2 - \left(\frac{u_{\max} - u_{\min}}{2} \right)^2 = 0, \quad (14)$$

where $u_{\min}$ and $u_{\max}$ are the minimum and maximum thrusts, respectively, that each rotor can generate. The performance index was

$$\begin{aligned}
J = & (x_d - x_{\text{ref}})^T S_f (x_d - x_{\text{ref}}) \\
& + \int_t^{t+T} \left((x_d - x_{\text{ref}})^T Q (x_d - x_{\text{ref}}) \right. \\
& \left. + \sum_{i=1}^6 r_i (u_i - u_{\text{ref}i})^2 - \sum_{i=1}^6 r'_i u'_i \right) d\tau,
\end{aligned} \quad (15)$$

where $u_i = f_i$ ($i = 1, \dots, 6$) are control inputs, $u_{\text{ref}i}$ ($i = 1, \dots, 6$) are reference values of inputs, $x_{\text{ref}} \in \mathbb{R}^{12}$ is a reference vector of state, $S_f, Q \in \mathbb{R}^{12 \times 12}$ are weighting matrices, and r_i, r'_i ($i = 1, \dots, 6$) are weighting values. The reference values of the inputs in each case are listed in Table 1, the weighting matrices of state are in Table 2, and the weighting values of inputs are in Table 3. The linear penalty for the dummy inputs in the performance index was introduced to guarantee the uniqueness of the solution. We chose these weights by trial and error. The terminal weights were set to be equal to the stage weights. In Case 1, in which three rotors in alternate positions stop, it was impossible to control the yaw angle because all of the remaining rotors rotated in the same direction. Therefore, the weight of the yaw angle was set to zero.

In Case 2, in which two adjacent rotors and a separate rotor stop, stabilization of the attitude was a top priority for maintaining altitude. We set up reference values of inputs to support the hexacopter by thrusts of two diagonal rotors (u_1 and u_4) and reduced the thrust of the remaining rotor, u_2 . Since the hexacopter was supposed to be able to support its own weight only by the thrusts of u_1 and u_4 , the input constraint was relaxed for the simulation study as follows:

$$u_{\min} = 0 \text{ [N]}, u_{\max} = 12 \text{ [N]}. \quad (16)$$

Although this relaxed constraint cannot be satisfied by Firefly, we can investigate the performance of NMPC for a generic hexacopter.

Finally, we considered Case 3, in which three adjacent rotors stop. In this case, since the thrust exists only on one side of the hexacopter, and it is impossible to maintain the attitude from a static state, we thought to maintain horizontal attitude and altitude by rotating in the yaw direction. Therefore, the reference value $\dot{\psi}_{\text{ref}} = 12 \text{ [rad/s]}$ was, in this case, only given for the component of yaw angular velocity as follows:

$$x_{\text{ref}} = (0 \ \dots \ 0 \ \dot{\psi}_{\text{ref}})^T \in \mathbb{R}^{12}. \quad (17)$$

We chose this reference by trial and error.

Table 1. Reference values of the inputs

	$u_{\text{ref}1}$	$u_{\text{ref}2}$	$u_{\text{ref}3}$	$u_{\text{ref}4}$	$u_{\text{ref}5}$	$u_{\text{ref}6}$
Case 1	$\frac{mg}{3}$	—	$\frac{mg}{3}$	—	$\frac{mg}{3}$	—
Case 2	$\frac{mg}{2}$	0	—	$\frac{mg}{2}$	—	—
Case 3	$\frac{mg}{3}$	$\frac{mg}{3}$	$\frac{mg}{3}$	—	—	—

Table 2. Weighting matrices of state

	S_f, Q
Case 1	diag[5 5 20 10 10 0 0.1 0.1 0.1 0.1 0.1 0]
Case 2	diag[1 1 15 10 10 0 0.1 0.1 0.1 0.1 0.1 0]
Case 3	diag[10 10 10 10 10 0 0.1 0.1 0.1 0.1 0.1 10]

4. SIMULATIONS

4.1 Simulation Conditions

The simulations were performed on a personal computer (CPU: Intel(R) Core(TM) i5-6200U 2.30 [GHz], RAM: 8.00 [GB], OS: Windows 7 Professional). The program in C language was written by an automatic code generation system called AutoGenU (Ohtsuka (2015)). AutoGenU is a tool from a symbolic computing environment, Maple, for generating a simulation program for NMPC.

The physical parameters of the hexacopter were: $m = 1.44 \text{ [kg]}$, $g = 9.81 \text{ [m/s}^2\text{]}$, $l = 0.23 \text{ [m]}$, $k = 1.60 \times 10^{-9} \text{ [m]}$, $J_{xx} = 0.0348 \text{ [kg} \cdot \text{m}^2\text{]}$, $J_{yy} = 0.0459 \text{ [kg} \cdot \text{m}^2\text{]}$, and $J_{zz} = 0.0977 \text{ [kg} \cdot \text{m}^2\text{]}$. These parameters were measured from a Firefly from AscTec. $\gamma = 1 \times 10^{-2} \text{ [N} \cdot \text{m} \cdot \text{s/rad]}$ was arbitrarily determined with reference to Muller et al. (2014).

4.2 Simulation Results

In all three cases, the optimal control problem in the finite horizon of $T = 1.0 \text{ [s]}$ was solved within 0.75 [ms] for the sampling period 1 [ms]. That is, real-time optimization was achieved. In addition, as shown below, NMPC was superior to the conventional linear control because the position and attitude could be controlled from an initial state that significantly deviated from the state of equilibrium or despite the absence of static equilibrium.

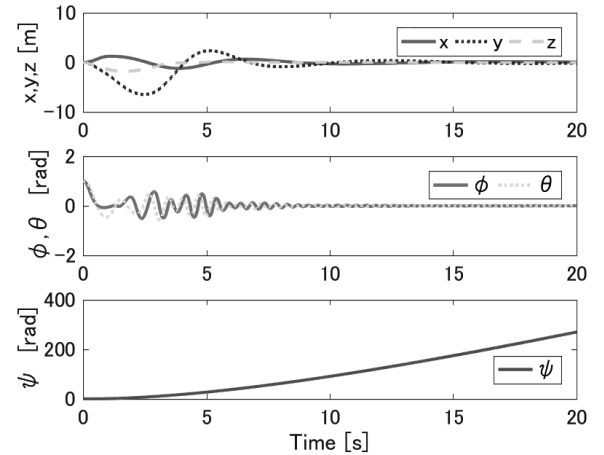


Fig. 5. Time histories of state variables in Case 1

The simulation results when three rotors stop are shown in Figs. 5 to 10. First, we considered Case 1, in which three rotors in alternate positions stopped. The initial conditions were given as $\xi_0 = (0 \ 0 \ 0)^T$, $\eta_0 = (1 \ 1 \ 0)^T$, $\dot{\xi}_0 = (0 \ 0 \ 0)^T$, and $\dot{\eta}_0 = (0 \ 0 \ 0)^T$. These initial conditions were determined by considering the delay in fault detection. As shown in Figs. 5 and 6, the position and attitude were controlled from an initial state that significantly deviated from the state of equilibrium, while in previous research (Hirai (2017)), the horizontal position was unable to be controlled unless it was from a state of near equilibrium. The rotation around the yaw axis increased as time passed because all three rotors rotated in the same direction and

Table 3. Weighting values of inputs

	r_1	r_2	r_3	r_4	r_5	r_6	r'_1	r'_2	r'_3	r'_4	r'_5	r'_6
Case 1	1	0	1	0	1	0	0.1	0	0.1	0	0.1	0
Case 2	1	1	0	1	0	0	0.1	0.1	0	0.1	0	0
Case 3	10	10	10	0	0	0	0.01	0.01	0.01	0	0	0

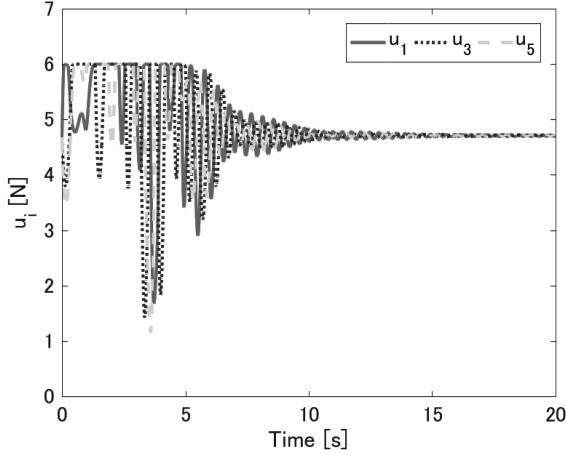


Fig. 6. Time histories of input variables in Case 1

the force around the yaw axis was large. After a certain period of time, the torque in the yaw direction became balanced with the air drag, and the yaw angular velocity $\dot{\psi}$ converged to a constant.

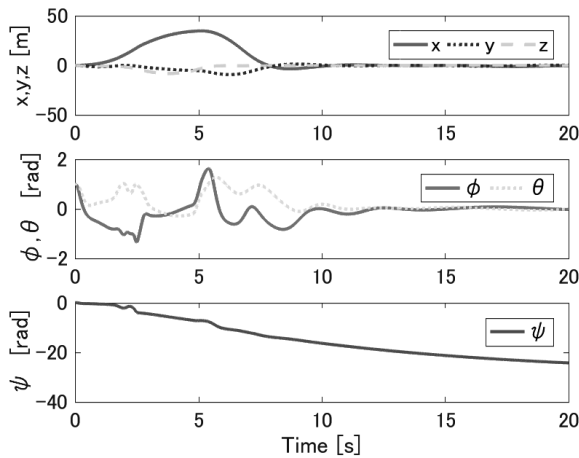


Fig. 7. Time histories of state variables in Case 2

Next, we considered Case 2, in which two adjacent rotors and a separate rotor stopped. The initial conditions were given as $\xi_0 = (0 \ 0 \ 0)^T$, $\eta_0 = (1 \ 1 \ 0)^T$, $\dot{\xi}_0 = (0 \ 0 \ 0)^T$, and $\dot{\eta}_0 = (0 \ 0 \ 0)^T$. These initial conditions were determined by considering the delay in fault detection. As shown in Figs. 7 and 8, the weight of the hexacopter was supported mainly by the thrusts of two diagonal rotors (u_1 and u_4), and its position and attitude were successfully controlled. In contrast, it had been impossible to maintain the altitude of the hexacopter by conventional linear control, which shows an advantage of NMPC because it explicitly takes nonlinear dynamics into account.

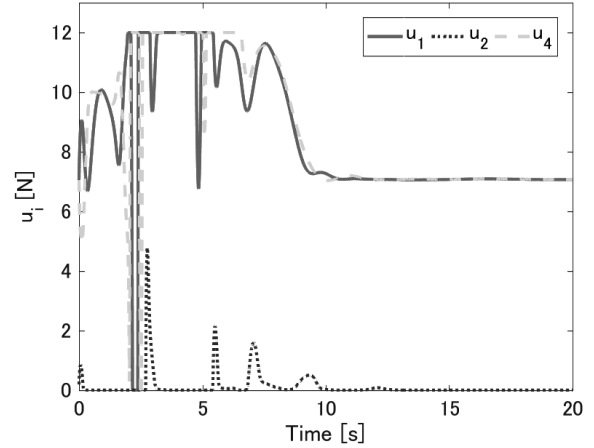


Fig. 8. Time histories of input variables in Case 2

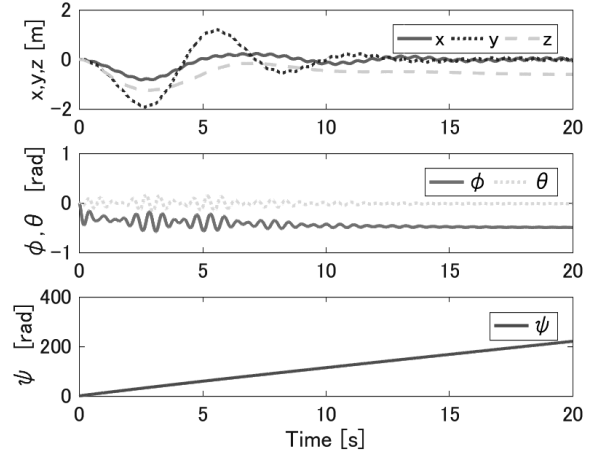


Fig. 9. Time histories of state variables in Case 3

Finally, we considered Case 3, in which three adjacent rotors stopped. Unlike the other cases, the initial conditions were given as $\xi_0 = (0 \ 0 \ 0)^T$, $\eta_0 = (0 \ 0 \ 0)^T$, $\dot{\xi}_0 = (0 \ 0 \ 0)^T$, and $\dot{\eta}_0 = (0 \ 0 \ 12)^T$. This initial yaw angular velocity was chosen through trial and error because there was no static equilibrium state and it was difficult to maintain the attitude without the yaw angular velocity. As shown in Figs. 9 and 10, we succeeded in simultaneously maintaining the altitude of the hexacopter and controlling the position in a horizontal direction. It should be noted that there was no need to find a nonzero equilibrium beforehand during the NMPC design. In Fig. 10, we can see the oscillation of the inputs. The frequency of this oscillation is the same as that of the yaw rotation.

5. CONCLUSION

In this research, we applied NMPC in order to control the position and attitude of a hexacopter with three

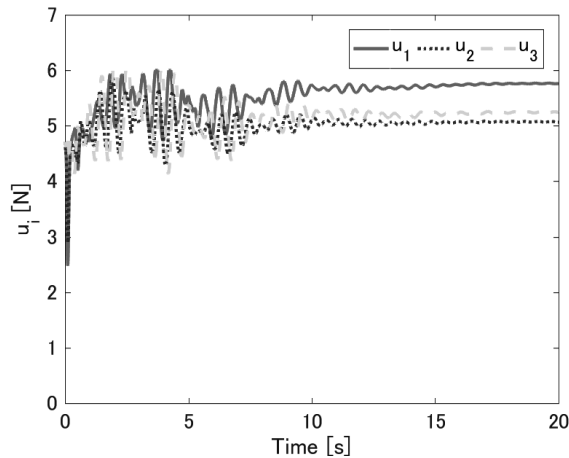


Fig. 10. Time histories of input variables in Case 3

failed rotors. As a result, it was possible to control the horizontal position in addition to maintaining altitude, whereas it was impossible to control the horizontal position in previous research using linear control (Hirai (2017)). In particular, the equilibrium in Case 2 could not be stabilized by any linear feedback control, and it was not a trivial task to find a nonzero equilibrium in Case 3. NMPC could handle all the cases with the same performance index and constraints on the control inputs. Future work includes controlling a hexacopter with four or five failed rotors, fault detection, considering the delay of motors, and a hardware experiment with an actual hexacopter to demonstrate real-time optimization of NMPC.

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